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Hybrid Water Based Absorption and Membrane for Deep Desulfurization of Claus Unit Tail Gas

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Abstract

A novel approach for the deep removal of sulfur dioxide (SO_2) from sulfur recovery unit (SRU) tail gas is presented. This new process combines an absorption process, where SO_2 is scrubbed/absorbed by mildly acidic water, followed by a membrane-based enrichment step to concentrate sulfur species before recycling them back to the Sulfur Recovery Unit (SRU).

SRUs convert hydrogen sulfide (H_2S) into elemental sulfur (S) by the modified Claus process with a yield that can reach up to 99% by partial oxidation with oxygen (O_2). To meet stringent SO_2 emissions standards, higher recovery rates are imperative. Achieving the required high sulfur recovery efficiency e.g. 99.9% or higher, necessitates the installation of a tail gas treatment. The prevailing technology for tail gas treatment is the reduction absorption, involving hydrogenation/hydrolysis step to convert all sulfur compounds/species exiting SRU into H_2S . This is followed by an absorption step with H_2S -selective amine base solvent to minimize carbon dioxide (CO_2) co-absorption. The amine solution is then regenerated through heating to release H_2S and co-absorbed CO_2 , which are recycled to SRU furnace reactor. However, this regeneration step is energy intensive and significantly contributes to the carbon footprint of the sulfur recovery process. Alternative processes with lower energy intensity, reduced carbon footprint as well as lower capital and operating costs are preferable.

In the novel approach, there is no hydrogenation step. Instead, the tail gas from the SRU is sent to a thermal oxidizer, which converts all sulfur compounds and species into SO_2 and SO_3 , collectively referred to as SO_x . Downstream of the thermal oxidizer, and heat recovery, SO_x are scrubbed from the tail gas by a mildly acidic water in a column at 60 °C. As water is produced during the H_2S conversion (i.e. H_2S combustion with O_2), no water makeup is required, except for the initial water load during the startup. The absorbed SO_x are concentrated using Reverse Osmosis and Nano-Filtration membranes, with the membrane rejecting SO_x and the concentrated stream sent back to the SRU. Alternatively, SO_x stream can be further concentrated to saleable sulfuric acid. The permeate of the membrane can be utilized for further scrubbing, directed to an evaporation pond, or reused within the plant.

Introduction

The global energy landscape is undergoing a significant transformation, driven by increasing demand for cleaner fuels and stricter environmental regulations. Within this context, the efficient and environmentally responsible processing of hydrocarbons, particularly natural gas and crude oil, has become paramount. A critical aspect of this processing involves the removal of sulfur compounds, primarily hydrogen sulfide (H_2S), which are naturally present in the raw oil and gas. H_2S is a highly toxic, corrosive, and flammable gas that poses significant health, safety, and environmental risks. Its direct release into the atmosphere is strictly prohibited due to its detrimental effects, including its contribution to acid rain formation and its direct toxicity to living organisms [1].

Sulfur Recovery Units (SRUs) [2] are therefore indispensable components of modern refineries, gas processing plants, and other industrial facilities that handle sulfur-containing feedstocks. The primary function of an SRU is to convert H_2S into elemental sulfur, a valuable commodity with diverse industrial applications. The most widely adopted technology for this purpose is the modified Claus process, which has been the industry standard for decades due to its robustness and high sulfur recovery efficiency, typically reaching up to 99% [3].

However, despite the high efficiency of the Claus process, a small but significant quantity of sulfur compounds remains in the treated gas stream, commonly referred to as tail gas. These residual sulfur species, including unreacted H_2S , sulfur dioxide (SO_2), carbonyl sulfide (COS), carbon disulfide (CS_2), and sulfur vapor are still present in concentrations that exceed stringent environmental emission limits. Consequently, further treatment of these tail gases is absolutely essential to comply with evolving environmental regulations and to minimize the overall environmental footprint of industrial operations. This necessity has led to the widespread adoption of Tail Gas Treatment Units (TGTUs) downstream of SRUs [4].

Traditional TGTUs, while effective, often come with their own set of challenges, including high energy consumption, significant capital and operating costs, and a considerable carbon footprint, particularly those relying on amine-based absorption processes that require energy-intensive regeneration steps. The continuous drive for more sustainable and economically viable solutions has encouraged innovation in this critical area of environmental engineering.

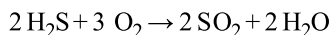
This research paper introduces a novel and promising approach for the deep desulfurization of Claus unit tail gas, combining water-based absorption with advanced membrane separation technologies. This hybrid process aims to overcome the limitations of conventional TGTUs by offering a more energy-efficient, environmentally friendly, and cost-effective solution. We will provide a comprehensive overview of the fundamental principles of the Claus process, a detailed analysis of the characteristics of tail gas streams, and an introduction to the prevalent tail gas treatment technology. The core of this paper will focus on the innovative hybrid water-based absorption and membrane process, detailing its operational stages, pilot validation, and its significant technical, economic, and environmental advantages. Furthermore, we will discuss the inherent technical challenges in tail gas treatment, and the solutions being developed to address them. Finally, we will explore future trends and developments in sulfur recovery and tail gas treatment, including the potential for third-party implementation and operation models.

The Claus Process and its Limits

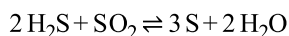
Fundamental Principles of the Modified Claus Process

The Claus process, named after Carl Friedrich Claus who developed it in the late 19th century [5], stands as the cornerstone of sulfur recovery from H_2S -rich gas streams in various industrial sectors, particularly oil refining and natural gas processing. Its primary objective is to convert hazardous H_2S into valuable elemental sulfur, thereby mitigating environmental concerns and yielding a marketable product. The process fundamentally involves a series of thermal and catalytic reactions.

The initial phase, known as the thermal stage, occurs in a reaction furnace at high temperatures, typically ranging from 1,000 to 1,400 °C. In this stage, approximately one-third of the H₂S in the acid gas feed is partially oxidized with air or oxygen-enriched air to produce sulfur dioxide (SO₂) and water vapor. This reaction is highly exothermic and provides the necessary heat for complete destruction of hydrocarbon and ammonia, if any, and subsequent reactions:



Following this, the newly formed SO₂ reacts with the remaining unreacted H₂S to produce elemental sulfur and water. This is the core Claus reaction:



After the thermal stage, the gas stream is cooled, and the elemental sulfur produced is condensed and removed. The gas then proceeds to the catalytic stage, which typically consists of two or three catalytic converters in series, each followed by a sulfur condenser. These converters house alumina-based catalysts that facilitate the Claus reaction at lower temperatures (200–350 °C), shifting the equilibrium towards further sulfur formation [6].

Sulfur Recovery Efficiency and Residual Emissions

The efficiency of the Claus unit in sulfur recovery is influenced by several factors, including the H₂S concentration in the feed gas, the type of catalyst employed, and the operating conditions. Higher H₂S concentrations generally lead to increased sulfur recovery efficiency. Furthermore, continuous advancements in catalyst technology, such as the development of more active and selective catalysts, have significantly contributed to enhancing the overall performance of Claus units. Conventional Claus plants typically achieve sulfur recovery efficiencies ranging from 95% to 97%, depending on the number of catalytic stages [7].

Despite these high recovery rates, a significant challenge remains: the presence of residual sulfur compounds in the tail gas. These emissions, though representing a small percentage of the total sulfur, are still substantial enough to necessitate further treatment to meet increasingly stringent environmental regulations. The primary residual sulfur species include unreacted H₂S, SO₂, elemental sulfur vapor and particularly problematic compounds like carbonyl sulfide (COS) and carbon disulfide (CS₂).

Formation of COS and CS₂

Carbonyl sulfide (COS) and carbon disulfide (CS₂) are undesirable byproducts that can form in the Claus reaction furnace, especially when hydrocarbons are present in the acid gas feed. These compounds are more stable than H₂S and SO₂ and do not react efficiently under typical Claus process conditions. Their formation pathways involve reactions between H₂S, CO₂, and hydrocarbons at high temperatures [8, 9]. For instance, COS can form from the reaction of H₂S with CO or CO₂. These compounds are problematic because they are not effectively converted to elemental sulfur in the catalytic stages of the Claus process and therefore pass through to the tail gas stream. This requires their conversion back to H₂S in the Tail Gas Treatment Unit (TGTU) through hydrogenation and hydrolysis reactions, adding complexity and cost to the overall sulfur recovery process [10].

Necessity of Tail Gas Treatment Units (TGTU)

The limitations of the Claus process, particularly its inability to achieve near-100% sulfur recovery and its inefficiency in converting COS and CS₂, underscore the critical need for TGTUs. These downstream units are designed to capture and convert the residual sulfur compounds in the Claus tail gas to meet stringent emission standards, often requiring overall sulfur recovery efficiencies of 99.9% or higher. Without TGTUs, the SO₂ and other sulfur species emitted from Claus plants would contribute significantly to

air pollution, leading to environmental degradation and potential regulatory non-compliance. Therefore, TGTUs are not merely an add-on but an integral and essential part of modern sulfur recovery facilities, ensuring environmental protection and sustainable industrial operations.

Most practiced Tail Gas Treatment Unit (TGTU) Technology

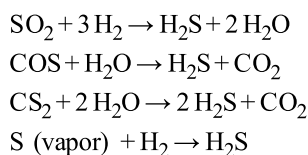
To achieve the stringent sulfur emission limits and overall sulfur recovery efficiencies exceeding 99.5%, Sulfur Recovery Units (SRUs) are invariably followed by Tail Gas Treatment Units (TGTUs). These units are designed to capture and convert the residual sulfur compounds (H_2S , SO_2 , COS , CS_2 , and elemental sulfur vapor) present in the Claus tail gas into a form that can be either recycled back to the SRU or safely disposed of. Various technologies have been developed and commercialized for this purpose, each with its own set of advantages, disadvantages, and specific applications.

Reduction/Absorption Processes (SCOT)

Reduction/absorption processes represent the most prevalent and widely adopted TGTU technology in the industry. These processes are characterized by a two-step approach: first, the catalytic reduction of all sulfur species to H_2S , followed by the selective absorption of H_2S . The most well-known and widely implemented process in this category is the Shell Claus Off-gas Treating (SCOT) process [11].

Detailed Description

In a typical reduction/absorption TGTU, the Claus tail gas, after being reheated, is mixed with a reducing gas (typically hydrogen, H_2 , and carbon monoxide, CO , produced by burning a small amount of fuel gas with insufficient air). This mixture is then passed through a catalytic reactor, usually containing a cobalt-molybdenum (CoMo) catalyst on an alumina support. In this hydrogenation/hydrolysis reactor, all sulfur compounds are converted into H_2S . The key reactions occurring are:



After the catalytic conversion, the gas stream, now rich in H_2S , is cooled and then routed to an absorption tower. Here, H_2S is selectively absorbed by a circulating aqueous amine solution, typically a tertiary amine like Methyldiethanolamine (MDEA) or a hindered amine. These amines are chosen for their selectivity towards H_2S over carbon dioxide (CO_2), which is also present in significant quantities in the tail gas. The selective absorption minimizes the co-absorption of CO_2 , which would otherwise reduce the capacity of the amine unit and increase the load on the Claus unit if recycled.

Once the amine solution is rich in H_2S , it is sent to a regenerator where, through heating (typically with low-pressure steam), the H_2S is stripped from the solution. This regenerated H_2S -rich acid gas stream is then recycled back to the Claus reaction furnace, effectively closing the sulfur recovery loop. The lean amine solution is cooled and returned to the absorber. The treated gas, now with very low sulfur content, is typically sent to an incinerator for final oxidation of any remaining trace sulfur compounds to SO_2 before being vented to the atmosphere [12].

Advantages and Disadvantages

Advantages:

- High Sulfur Recovery: Reduction/absorption processes can achieve very high overall sulfur recovery efficiencies, often exceeding 99.9%, enabling compliance with the most stringent emission regulations.

- Proven Technology: These processes have a long and successful track record of operation in various industrial settings, making them a reliable choice.
- Flexibility: They can handle variations in tail gas composition and flow rates to some extent.

Disadvantages:

- High Energy Consumption: The amine regeneration step is highly energy-intensive, requiring significant amounts of steam for heating. This contributes to high operating costs and a substantial carbon footprint for the overall sulfur recovery process.
- Significant Footprint: These units typically require a large physical footprint due to the multiple vessels, heat exchangers, and associated piping.
- Capital and Operating Costs: The investment costs for reduction/absorption TGTUs are considerable, and the operating costs are high due to energy consumption and chemical makeup.
- CO₂ Co-absorption: While selective amines are used, some CO₂ is inevitably co-absorbed, especially in cases of lean acid gas feeds. This co-absorbed CO₂ is recycled to the Claus furnace, which can reduce the capacity and efficiency of the Claus unit itself.
- COS/CS₂ Conversion: Although the hydrogenation reactor is designed to convert COS and CS₂ to H₂S, incomplete conversion can occur. Unconverted COS, in particular, is poorly absorbed by H₂S-selective amines and can pass through the absorber, leading to higher SO₂ emissions after incineration [13, 14].

The Novel Hybrid Approach: Water-based Absorption and Membrane

Recognizing the limitations of conventional Tail Gas Treatment Units (TGTUs), particularly their high cost and significant energy consumption and carbon footprint, a novel hybrid approach has been developed for the deep desulfurization of Claus unit tail gas. This innovative process integrates water-based absorption with advanced membrane separation technologies, offering a more sustainable and economically viable alternative. Unlike traditional reduction/absorption processes, this new method eliminates the hydrogenation step and the need for energy-intensive amine regeneration, significantly reducing operational costs and environmental impact.

Overview of the Innovative Process

The proposed hybrid process fundamentally re-engineers the approach to tail gas treatment. Instead of reducing all sulfur species to H₂S for subsequent amine absorption, this technology first oxidizes all sulfur compounds to SO_x. These SO_x are then absorbed by a mildly acidic water solution, and the resulting sulfuric acid solution is concentrated using membrane technology. The concentrated sulfuric acid can then be recycled back to the Claus furnace, achieving a near-zero waste sulfur recovery process. As an alternative, concentrated sulfuric acid could be supplied to support the expansion of Saudi Arabia's mining sector. This streamlined approach offers several advantages, including lower energy intensity, reduced carbon footprint, and potentially lower capital and operating costs [15].

The process can be systematically divided into five elemental steps, each playing a crucial role in the overall efficiency and sustainability of the system:

- Thermal Oxidation of the Tail Gas: All sulfur species in the Claus tail gas are converted into SO₂.
- SO₂/SO₃ Gas Absorption in Mildly Acidic Solution: The oxidized gas is cooled, and SO₂ and SO₃ are absorbed by a circulating mildly acidic water solution.
- SO₂ Oxidation with Air: Dissolved SO₂ is further oxidized to sulfuric acid within the aqueous phase by injection of air.

oxidation step simplifies the downstream treatment, as all sulfur is now in the form of SO_2 , which is amenable to water-based absorption.

SO_2/SO_3 Gas Absorption in Mildly Acidic Solution. Following thermal oxidation, the hot gas stream is cooled in a waste heat boiler, recovering valuable energy, and then further cooled in a quench tower. This cooling process facilitates the simultaneous partial condensation of water vapor and the absorption of SO_2 in a counter-current flow of mildly acidic water. The absorption occurs in a column designed for efficient gas-liquid contact.

Sulfur dioxide (SO_2) exhibits significant solubility in water, and this solubility is influenced by temperature and the acidity of the solution. As shown in Figure 2, SO_2 solubility in water and aqueous solutions of various sulfuric acid concentrations varies with temperature. In the targeted operating conditions for this process, with an acid content of approximately 0.1 wt% sulfuric acid and temperatures ranging from 60 to 80 °C, the SO_2 solubility ranges from 20 g/kg at 80 °C to 30 g/kg at 60 °C. This temperature range is chosen to optimize both SO_2 absorption efficiency and subsequent membrane performance.

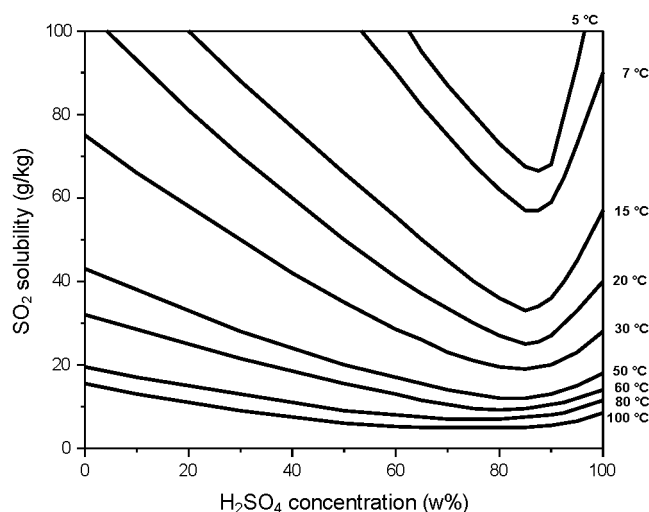
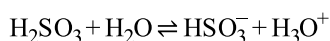
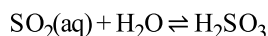
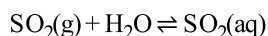


Figure 2— SO_2 Solubility in several sulfuric acid concentration at various temperature (°C) and at 1 atmosphere of SO_2 derived from reference [21].

Upon dissolution in water, gaseous SO_2 ($\text{SO}_2(\text{g})$) can either physically dissolve to form aqueous SO_2 ($\text{SO}_2(\text{aq})$) or undergo hydration to yield sulfurous acid (H_2SO_3) [20]. Sulfurous acid is a weak acid that dissociates in water to generate bisulfite ions HSO_3^- and hydronium ions H_3O^+ :



In contrast, sulfur trioxide ($\text{SO}_3(\text{g})$), if present, undergoes a rapid and complete hydration reaction to produce sulfuric acid (H_2SO_4), a strong acid that fully dissociates in water to form bisulfate (HSO_4^-) and sulfate (SO_4^{2-}) ions, along with hydronium ions (H_3O^+) [21]:

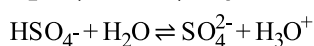
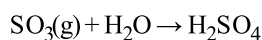


Figure 3 illustrates the speciation of sulfur compounds in aqueous solution as a function of pH. At pH values between 1.8 and 3, which are typical for the mildly acidic absorption solution, most sulfur species

are present as HSO_3^- and SO_4^{2-} . Carbon compounds, primarily from co-absorbed CO_2 , remain in the form of aqueous CO_2 . The presence of a small amount of H_2SO_4 (e.g., 0.1 wt%) contributes to the overall acidity of the solution. The continuous production of water during the H_2S combustion step means that, after initial startup, no external water makeup is required for the absorption process, making it self-sufficient in terms of water balance.

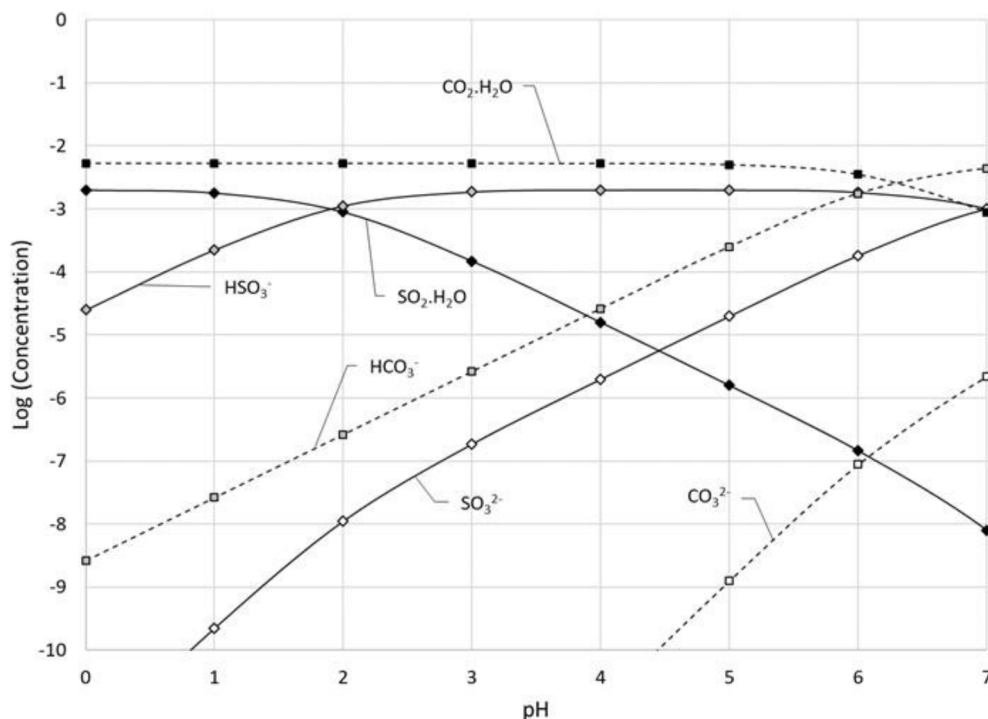


Figure 3—pH dependence of aqueous sulfur dioxide and carbon dioxide speciation in equilibrium with a gas phase containing 36 kPa of CO_2 and 0.4 kPa of SO_2 at 60°C in a 0.1 w% H_2SO_4 solution.

SO_2 Oxidation with O_2 in Solution. A critical aspect of this hybrid process is the in-situ oxidation of dissolved SO_2 within the aqueous phase. As soon as SO_2 dissolves in the water, it can be oxidized by the residual oxygen (O_2) present in the gas stream, leading to the formation of sulfuric acid. This reaction can occur via a fast electrochemical process.

Research by Lancia et al. has investigated the kinetics of these redox reactions [22]. Their studies indicated that the reaction rates were zero-order with respect to dissolved oxygen and 3/2-th order for the bisulfite ion (HSO_3^-), which originates from the dissociation of dissolved $\text{SO}_2(\text{aq})$ and H_2SO_3 . This spontaneous oxidation of $\text{SO}_2(\text{aq})$ to $\text{SO}_3(\text{aq})$ or H_2SO_4 within the absorption solution contributes to the overall efficiency of sulfur capture and conversion, reducing the load on subsequent treatment steps. The complete oxidation can be carried out in a bubble column fed with pressurized air.

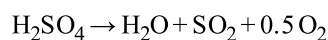
H_2SO_4 Concentration by Membranes (NF/RO). Following the absorption and in-situ oxidation steps, the diluted sulfuric acid stream needs to be concentrated for efficient recycling or production of saleable quality sulfuric acid. This is achieved using advanced membrane technologies, specifically Nanofiltration (NF) and Reverse Osmosis (RO) membranes, or a combination thereof. Electrodialysis is also a viable option for this concentration step. These membrane processes offer a cost-effective and environmentally friendly alternative for pre-enriching diluted mildly acidic sulfuric acid solutions compared to traditional evaporation or distillation methods, which are highly energy-intensive.

RO and NF membranes exploit the principle of selective permeability to separate solutes based on their size, charge, and interaction with the membrane surface. RO membranes operate at high pressures, forcing

water molecules through the membrane while effectively retaining larger solutes, including sulfuric acid. NF membranes, requiring lower operating pressures than RO, provide an intermediate solution between RO and ultrafiltration, making them suitable for applications where a partial separation of ions is desired. In this context, the membranes are designed to reject the H_2SO_4 species (sulfuric acid) while allowing water to permeate through, thus concentrating the sulfuric acid solution [23, 24].

Commercial membrane modules are utilized for this concentration step. An example of sulfuric acid enrichment using an RO membrane is provided in Table 2. The data demonstrates that in a single step, the sulfuric acid concentration can be effectively doubled, and crucially, the permeate stream exhibits a very low sulfuric acid content. This clean permeate, essentially fresh water, can be recycled back as a solvent for SO_2 absorption, directed to an evaporation pond, or reused elsewhere within the plant, further enhancing the process's sustainability and avoiding any water makeup requirements. The concentrated sulfuric acid stream, typically ranging from 10-20 wt.% H_2SO_4 , is then ready for the final step of the process.

H_2SO_4 Combustion in the Claus Furnace. The final step in this hybrid process involves sending the concentrated sulfuric acid stream back to the Claus reaction furnace for combustion. This step is pivotal in achieving a near-zero waste sulfur recovery process. In the high- temperature environment of the Claus furnace, the sulfuric acid (H_2SO_4) undergoes thermal decomposition, breaking down into water (H_2O), sulfur dioxide (SO_2), and oxygen (O_2) [25]:



This decomposition effectively recycles the captured sulfur back into the main Claus process in the form of SO_2 , which can then react with H_2S to produce elemental sulfur. A significant advantage of this approach is that the decomposition of H_2SO_4 also releases oxygen (O_2). This internally generated oxygen can reduce the demand for external air supply to the Claus furnace, potentially leading to energy savings and a more efficient combustion process. Furthermore, by recycling the sulfuric acid, the process avoids the generation of a liquid waste stream, contributing to a more environmentally friendly operation. While there is a potential effect on the reaction furnace temperature due to the endothermic decomposition of H_2SO_4 , this needs to be assessed on a case-by-case basis depending on the specific Claus unit design and operating conditions. Overall, this combustion step closes the loop, ensuring that all captured sulfur is effectively recovered, and the process operates with minimal environmental discharge.

Pilot Tests and Experimental Results

To validate the feasibility and performance of the novel hybrid water-based absorption and membrane process, a series of pilot-scale tests were conducted. These experiments focused on two critical aspects of the proposed technology: the efficiency of SO_2 absorption in mildly acidic water and the effectiveness of membrane technology for regenerating and concentrating the acidic solution.

Absorption Pilot Set-up

A dedicated pilot-scale absorption system was constructed to gather proprietary data on the absorption of SO_2 by mildly acidic water. The setup comprised a 30-cm internal diameter glass column with a packing height of 2.5 meters. This column was designed to simulate the counter-current absorption process envisioned for the industrial application. The experimental data collection included systematic variations in several key parameters to understand their impact on SO_2 absorption efficiency:

- SO_2 content in the gas stream: Simulating different concentrations of SO_2 in the tail gas.
- Absorbent pH: Investigating the effect of varying acidity levels of the water solution.
- Liquid/Gas ratio: Optimizing the flow rates of the absorbent liquid and the gas stream.

- Kinetics of SO₂ oxidation in mildly acidic water: Studying the rate at which dissolved SO₂ is oxidized to sulfuric acid within the absorption column.

The experimental setup allowed for operation with liquid flow rates ranging from 0.5 to 1.35 m³/hr and gas flow rates between 50 to 100 Nm³/hr. The data obtained from these tests provided crucial insights into the optimal operating conditions for the absorption stage and confirmed the effectiveness of mildly acidic water as an absorbent for SO₂. Table 1 provides the experimental results for the absorption of SO₂ with water, along with the correlation from a model developed using commercial simulation software. This correlation enables the extrapolation of pilot-scale data to a demonstration-scale unit.

Table 1—Main process parameters for evaluating mildly acidic water absorption of SO₂ from air at 45°C and different water flowrates. Good correlations have been found between the experimental data and the process simulation

Run		AB-1	AB-2	AB-3	AB-4
Air flow in	m ³ /hr	50	50	50	50
SO ₂ inlet (GC)	ppm	6,100	6,100	5,790	5,790
SO ₂ outlet (GC)	ppm	60	99	464	464
Water flow in	m ³ /hr	1.35	1.35	1.15	1.15
SO ₂ outlet predicted	ppm	80	80	332	332

Mildly Acidic Water Regeneration with Membrane

Following the absorption step, the regeneration and concentration of the mildly acidic water solution are critical for the economic viability and sustainability of the process. Commercial Reverse Osmosis (RO) and Nanofiltration (NF) membranes were assessed for this purpose. Membrane technology, operating in the liquid phase, offers a cost-effective and environmentally friendly alternative for concentrating mildly sulfuric acidic solutions compared to traditional thermal separation methods.

RO and NF membranes operate on the principle of selective permeability, separating solutes based on their size, charge, and interaction with the membrane surface. RO membranes, requiring high operating pressures, force water molecules through the membrane while effectively retaining larger solutes, including sulfuric acid. NF membranes, on the other hand, operate at lower pressures than RO and provide an intermediate solution between RO and ultrafiltration, making them suitable for applications where a partial separation of ions is desired.

An example of sulfuric acid enrichment using an RO membrane is presented in Table 2. This table details the composition of the permeate of an RO membrane used to concentrate mildly acidic sulfuric aqueous solutions at 30 °C and different operating pressures. The results are highly promising:

Table 2—Composition of permeate of RO membrane to concentrate mildly acid sulfuric aqueous solutions at 30°C and different operating pressures.

Operating Pressure (bar)	Feed Flow (m ³ /hr)	Feed Sulfuric Acid (g/L)	Permeate Flow (m ³ /hr)	Permeate Sulfuric Acid (g/L)	Reject Flow (m ³ /hr)	Reject Sulfuric Acid (g/L)
5	1.00	1.28	0.15	0.04	0.85	1.47
15	1.00	1.28	0.39	0.03	0.61	2.06
25	1.00	1.28	0.53	0.03	0.47	2.84
30	1.00	1.28	0.58	0.03	0.42	3.14

As evident from Table 2, in a single step, the concentration of sulfuric acid can be effectively doubled (e.g., from 1.28 g/L in the feed to 3.14 g/L in the reject at 30 bar). Crucially, the permeate stream consistently exhibits a very low sulfuric acid content (as low as 0.03 g/L), indicating efficient separation. This clean

permeate can be directly recycled as a solvent for SO_2 absorption, significantly reducing water consumption and operational costs. These pilot results demonstrate the high potential of membrane technology for efficient and cost-effective regeneration and concentration of the acidic absorbent solution, paving the way for a more sustainable tail gas treatment process.

Technical, Economic, and Environmental Advantages of the Hybrid Approach

The novel hybrid water-based absorption and membrane process for deep desulfurization of Claus unit tail gas offers a compelling suite of advantages over conventional Tail Gas Treatment Units (TGTUs), particularly the widely used reduction/absorption processes. These benefits span technical performance, economic viability, and environmental sustainability, positioning this innovative approach as a promising solution for future sulfur recovery operations.

Technical Advantages

Elimination of Energy-Intensive Amine Regeneration. One of the most significant technical advantages of this hybrid process is the complete elimination of the energy-intensive amine regeneration step. In traditional reduction/absorption TGTUs, the amine solution, after absorbing H_2S , must be heated to high temperatures to strip the H_2S for recycling. This regeneration process consumes a substantial amount of low-pressure steam, contributing significantly to the overall energy demand and operating costs of the sulfur recovery complex. By replacing amine absorption with water-based absorption and membrane concentration, this process bypasses the need for this energy-intensive step, leading to substantial energy savings and a simplified process flow [26]. The new process can be achieve further decarbonization by using low-carbon intensity electricity.

Effective Treatment of COS and CS₂. Unlike conventional amine-based TGTUs where carbonyl sulfide (COS) and carbon disulfide (CS_2) can be problematic due to incomplete hydrolysis in the hydrogenation reactor and poor absorption by selective amines, the hybrid process effectively addresses these challenging sulfur species. In this new approach, all sulfur compounds, including COS and CS_2 , are first converted to SO_2 in a thermal oxidizer. This ensures that these compounds are fully oxidized to a form that is readily absorbed by the mildly acidic water solution, eliminating the issues associated with their incomplete conversion and subsequent emission [27]. This robust pre-treatment step simplifies the overall sulfur removal process and enhances the reliability of the TGTU. It is worth noting that the H_2S collected from sulfur pit air sweeping can be also converted to SO_2 and recovered.

Reduced CO₂ Co-absorption. Another notable technical benefit is the significant reduction in carbon dioxide (CO_2) co-absorption. Amine solutions, even those designed for H_2S selectivity, inevitably absorb some CO_2 , especially when the acid gas feed is lean in H_2S . This co-absorbed CO_2 is then recycled back to the Claus furnace, increasing the inert gas load and potentially reducing the capacity and efficiency of the Claus unit. The water-based absorption system, operating with mildly acidic water, exhibits a much lower affinity for CO_2 compared to amine solutions. This minimizes CO_2 co-absorption, leading to a cleaner recycle stream to the Claus unit and improved overall process efficiency [28].

Complete Avoidance of Heat-Stable Salt Issues. An advantage of the proposed hybrid desulfurization process is the complete avoidance of issues related to Heat-Stable Salts (HSS). In traditional amine-based "sweetening" plants, HSS are formed when strong acid anions react with the amine solution. These salts cannot be regenerated by heat, leading to a series of significant operational problems. The accumulation of HSS irreversibly reduces the active amine available for acid gas removal, thus lowering the system's overall efficiency. More critically, HSS are a primary cause of severe corrosion in key equipment like

reboilers and regenerator columns, which increases maintenance costs and can lead to unplanned shutdowns. Their presence also promotes operational issues such as foaming, fouling, and increased solution viscosity. By utilizing a non-amine, hybrid approach, the chemical pathway for HSS formation is removed. This inherently prevents solvent degradation, eliminates corrosion linked to HSS, and ensures sustained plant performance and reliability without the need for costly solvent reclaiming or replacement.

Zero-Waste Sulfur Recovery (H_2SO_4 Recycling). The ability to recycle the concentrated sulfuric acid back to the Claus furnace for decomposition is a key technical advantage that enables a near-zero waste sulfur recovery process. This closed-loop approach ensures that all captured sulfur is returned to the main process, eliminating the need for off-site disposal of spent absorbents or byproducts. The decomposition of H_2SO_4 in the furnace not only recovers SO_2 for further sulfur production but also releases oxygen, which can partially offset the need for external air supply, further enhancing the process's sustainability and resource efficiency [29].

Economic Advantages

Lower Operating Costs. The most direct economic advantage stems from the reduced energy consumption due to the elimination of amine regeneration. This translates into significant savings in steam and fuel gas costs. Furthermore, the reduced need for chemical makeup (amines) and the elimination of waste disposal costs for spent absorbents contribute to a lower overall operating expenditure. The self-sufficient water balance, where water produced during H_2S combustion is reused, also minimizes water utility costs.

Potentially Lower Capital Costs. While a detailed economic analysis would be required for specific projects, the simplified process flow and the elimination of complex amine regeneration and solvent storage systems could lead to potentially lower capital investment costs. The compact nature of membrane units compared to large absorption towers and regenerators might also contribute to a smaller footprint, reducing land requirements and associated civil engineering costs. The membrane unit can be installed at a distance from the SRU, as the mildly acidic water can be transported through piping without pressure loss constraints. The use of water as an absorbent further reduces initial chemical inventory costs.

Enhanced Process Efficiency and Capacity. By reducing CO_2 co-absorption and providing a cleaner recycle stream to the Claus unit, the hybrid process can indirectly enhance the efficiency and effective capacity of the existing SRU. This means that for a given SRU, the new TGTU could allow for higher throughput or better performance without requiring significant modifications to the main Claus plant, offering a cost-effective debottlenecking solution.

Environmental Advantages

Reduced Carbon Footprint. The elimination of energy-intensive amine regeneration directly translates into a significant reduction in the carbon footprint of the sulfur recovery process. Lower energy consumption means reduced emissions of greenhouse gases associated with energy generation. This aligns with global efforts to decarbonize industrial operations and meet climate change targets.

Compliance with Stringent Emission Standards. The hybrid process is designed to achieve deep desulfurization, ensuring that the treated tail gas meets and potentially exceeds the most stringent environmental emission standards for SO_2 . By effectively converting all sulfur species to SO_2 and then efficiently absorbing and recycling it, the process minimizes atmospheric releases of harmful sulfur compounds, protecting air quality and public health. The near-zero waste aspect further enhances its environmental performance by eliminating liquid waste streams associated with conventional TGTUs.

Resource Efficiency and Sustainability. By recycling the concentrated sulfuric acid and reusing the process water, the hybrid approach promotes resource efficiency and aligns with principles of the circular economy.

This minimizes the consumption of fresh resources and the generation of waste, contributing to the long-term sustainability of sulfur recovery operations. The inherent robustness in handling COS and CS₂ also means fewer upsets and more consistent environmental performance.

Environmental Regulations and Compliance

The global landscape of environmental regulations governing industrial emissions, particularly those from the oil and gas sector, has become increasingly stringent over the past few decades. This heightened regulatory scrutiny is a direct response to growing public awareness of environmental issues and the scientific understanding of the adverse impacts of pollutants on air quality, ecosystems, and human health. For Sulfur Recovery Units (SRUs) and their associated Tail Gas Treatment Units (TGTUs), compliance with these regulations is not merely a legal obligation but a fundamental aspect of operational license and corporate social responsibility. The primary pollutant of concern from SRU tail gas streams is sulfur dioxide (SO₂), which is subject to strict emission limits worldwide [30–35].

As far as Saudi Arabia is concerned, environmental regulations on SO₂ emissions are set by the Ministry of Environment Water and Agriculture (MEWA), formerly the General Authority of Meteorology and Environmental Protection (GAMEP). Prior to 2005, the standard required an SRU recovery rate greater than 95.0%, which could be readily achieved with two catalytic converters. After 2005, the requirement was tightened to a 98-99%, typically met through the addition of a third catalytic vessel and technologies such as SuperClaus®. A major update in 2014 further increase the requirement for gas plants to 99.9% SRU recovery, necessitating the addition of a Tail Gas Treatment (TGT) units. Most recently, in September 2021, the Air Quality Rule of Implementation (ROI) was issued. This regulation sets a maximum point-source limit of 250 ppmv (dry basis) and requires air dispersion modeling to establish site specific SO₂ emission limits, thereby defining the required SRU recovery efficiency on a case-by-case basis.

This evolving regulatory landscape has a profound and direct impact on the design, operation, and modernization of SRUs and TGTUs. To meet stricter emission limits, companies are compelled to invest in more advanced and efficient technologies. This has been a primary driver for the development of advanced Claus process variants (e.g., SuperClaus®) and highly efficient TGTUs capable of achieving overall sulfur recovery rates of 99.9% or even higher [36].

Design considerations are significantly influenced by these regulations. For instance, the choice of catalysts, the number of catalytic stages in the Claus unit, and the specific TGTU technology are all dictated by the required emission performance. Facilities in highly regulated areas or those with significant H₂S loads may opt for more complex and capital-intensive TGTUs to ensure compliance. Furthermore, the design must account for potential fluctuations in feed gas composition and flow rates, ensuring that the unit can maintain compliance under a variety of operating conditions.

Operationally, regulatory compliance mandates rigorous monitoring and control. Continuous Emission Monitoring Systems (CEMS) are often required for SRUs and TGTUs to provide real-time data on SO₂ and other pollutant emissions. These systems ensure ongoing compliance and provide data for regulatory reporting. Non-compliance can lead to severe penalties, including substantial fines, operational restrictions, and legal actions, in addition to reputational damage. Therefore, robust process control, regular maintenance, and proactive troubleshooting are essential to ensure consistent environmental performance. The need for high on-stream factors and reliability in TGTUs is also critical, as any downtime can lead to direct venting of untreated tail gas, resulting in immediate noncompliance and potential environmental harm.

In essence, environmental regulations have transformed from mere compliance checkboxes into fundamental drivers of technological innovation and operational excellence in the sulfur recovery industry.

Technical Challenges and Operational Considerations

The implementation and operation of any advanced industrial process, including the novel hybrid water-based absorption and membrane system for deep desulfurization, invariably present a unique set of technical challenges and operational considerations. While the proposed technology offers significant advantages, anticipating and addressing these potential hurdles is crucial for successful deployment and long-term performance.

Variability of Feed Gas

Sulfur Recovery Units (SRUs) often receive acid gas feeds whose composition (e.g., H_2S , CO_2 , hydrocarbon content) and flow rates can fluctuate significantly due to upstream process variations. These fluctuations can impact the performance of the thermal oxidizer, potentially leading to incomplete oxidation of sulfur species if not properly managed. While the thermal oxidizer is designed to handle a wide range of conditions, extreme or rapid changes in feed composition could necessitate dynamic control strategies to maintain optimal combustion and ensure complete conversion of all sulfur compounds to SO_2 . This requires robust online analytical capabilities and advanced process control systems to adjust combustion air and temperature in real-time [37].

Figure 4 illustrates the significant influence of temperature, the liquid-to-gas (L/G) molar ratio, and the initial SO_2 concentration on the efficiency of gas scrubbing with mildly acidic water. This is based on simulation work.

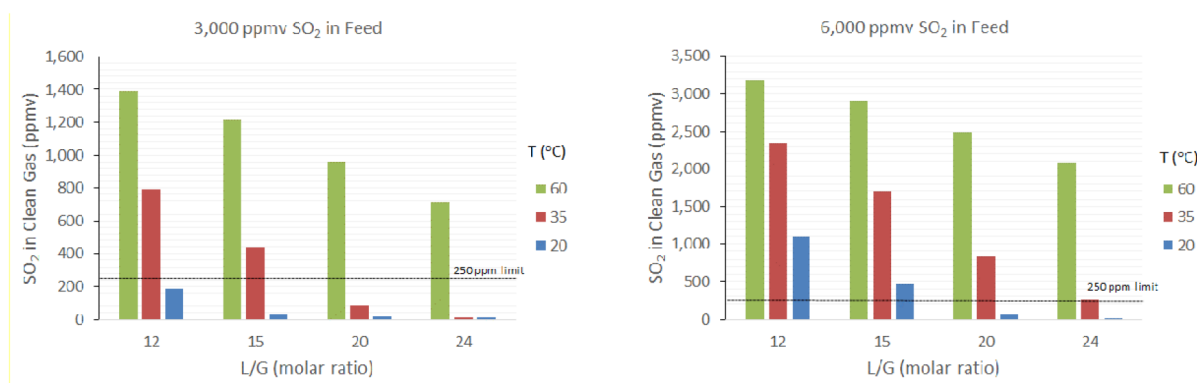


Figure 4—Dependence of SO_2 emissions on the liquid-to-gas molar ratio and operating temperature for very dilute sulfuric acid solution (0.01 wt%).

The first graph of Figure 4, corresponding to an initial SO_2 concentration of 3,000 ppmv in the feed gas, demonstrates a negative correlation between the operating temperature and absorption efficiency. At all simulated L/G ratios, an increase in temperature from 20°C to 60°C results in a higher residual concentration of SO_2 in the clean gas. This effect is particularly pronounced at 60°C, where SO_2 levels remain very high, far exceeding the regulatory limit of 250 ppmv. Conversely, a lower temperature, especially 20°C, significantly promotes SO_2 absorption, as the gas's solubility in mildly acidic water is greater (see Figure 2). Furthermore, increasing the L/G ratio from 12 to 24 systematically improves SO_2 removal performance. At 20°C, it is possible to meet the 250 ppmv limit with an L/G ratio of 12 and to perform even better with higher ratios. The second graph, where the initial SO_2 concentration is doubled to 6,000 ppmv, confirms the previously observed trends but underscores a greater load on the scrubbing system. The overall performance is logically reduced due to the larger amount of SO_2 to be treated. Although the impact of temperature and the L/G ratio follows the same logic, the conditions required to achieve regulatory compliance become more stringent. For example, at 35°C, an L/G ratio of 24 is needed to just get below the 250 ppmv threshold, whereas this condition was already met at an L/G ratio of 15 in the 3,000 ppmv scenario. At 20°C, an L/G ratio of 20 is

sufficient to achieve near-total SO₂ removal, demonstrating that low-temperature conditions and a high L/G ratio are crucial for treating gas streams with high SO₂ concentrations. In conclusion, these simulations indicate that for the effective removal of SO₂ using mildly acidic water, it is imperative to operate at a low temperature. Increasing the L/G ratio is also an effective strategy for improving performance, particularly when the initial SO₂ concentration in the feed gas is high

As mentioned above, lower operating temperatures increase SO₂ removal efficiency by mildly acidic water. However, achieving lower temperatures incurs additional cooling costs. In the Middle East, cooling to around 60 °C can be achieved with air coolers; below this threshold, chillers (e.g., propane-based) are required, which significantly increases the process capital and operating costs as well as its carbon footprint.

The novel process design must account for variability in feed gas composition and flowrate. For example, during turndown operation in n+1 mode, when one train is shut down for a planned turnaround and inspection, flowrate variations of up to 20% may occur. Such fluctuations affect absorber hydraulics and the quantity of SO₂ to be scrubbed. Increased flow may reduce sulfur recovery efficiency, leading to higher SO₂ concentrations in the tail gas. Catalyst degradation can also increase SO₂ emissions. The tail gas system should therefore be capable of handling such variations to a reasonable extent. If the malfunction is too severe or persists too long, the unit must be shut down for corrective action before resuming nominal operation.

The unit should be sized to balance capital and operating costs while ensuring operational flexibility. To support this, the technology includes features that enhance resilience against malfunctions. One option is the injection of an oxidizing compound, which provides additional SO₂ scrubbing capacity on demand. Another feature is a trim cooler installed on the absorbent line downstream of the air cooler and upstream of the absorption column. As illustrated in [Figure 3](#), lowering the temperature significantly enhances SO₂ absorption and thus helps manage temporary upsets. The availability of chilled water or propane refrigeration must, of course, be ensured on site.

Finally and unlike amine-based TGTUs, this novel tail gas treatment technology is unaffected by CO₂ content in the tail gas, since CO₂ has negligible solubility in mildly acidic water. Therefore, variations in feed gas CO₂ content do not translate into fluctuations in the treated gas.

Catalyst Fouling and Deactivation (in SRUs)

Although the novel hybrid process primarily focuses on tail gas treatment, its integration with the main Claus unit means that challenges within the SRU itself can indirectly affect the TGTU. Catalysts in the Claus reactors can suffer from fouling (e.g., by elemental sulfur deposition) or deactivation (e.g., by coking from hydrocarbons or poisoning by trace contaminants). While the new TGTU design aims to reduce the load on the Claus unit by recycling concentrated SO₂, maintaining the health and efficiency of the Claus catalysts remains paramount for the overall sulfur recovery performance. Regular monitoring, effective regeneration strategies, and potentially the use of more robust catalysts are necessary to mitigate these issues [38].

Precise Stoichiometry Control

For the overall Claus process, including the recycled SO₂ from the TGTU, maintaining a precise H₂S/SO₂ ratio (ideally 2:1) is critical for maximizing sulfur conversion. Any imbalance can lead to suboptimal recovery and increased emissions. With the introduction of recycled SO₂ from the TGTU, the control system for the Claus furnace needs to be sophisticated enough to account for this additional SO₂ load and adjust the primary air flow accordingly. This requires accurate measurement of both the incoming acid gas and the recycled stream, coupled with advanced process control algorithms to ensure optimal stoichiometry [39].

Management of Molten Sulfur

Elemental sulfur produced in the Claus process is in liquid form and must be maintained above its melting point (typically around 115-120 °C) to prevent solidification and blockages in pipelines and equipment. While this is a general challenge for all SRUs, the integration of the new TGTU, particularly the recycling of concentrated sulfuric acid, requires careful consideration of temperature management throughout the system. Ensuring that all lines handling liquid sulfur or concentrated acid are properly heated and insulated is crucial to prevent operational disruptions.

Durability and Long-Term Performance of Membranes

The long-term durability and performance of the Reverse Osmosis (RO) and Nanofiltration (NF) membranes are critical for the success of this hybrid process. Membranes can be susceptible to fouling, scaling, and chemical degradation, which can reduce their permeability and selectivity over time. Fouling can occur from suspended solids, organic matter, or inorganic precipitates in the mildly acidic water solution. Chemical degradation can be caused by exposure to harsh chemicals or extreme pH conditions.

To ensure long-term performance, robust pre-treatment of the absorbent solution (e.g., filtration to remove particulates), regular membrane cleaning protocols (e.g., chemical cleaning, backwashing), and careful selection of membrane materials resistant to the specific operating conditions (acidic environment, temperature) are essential. Continuous monitoring of membrane performance (e.g., flux, rejection rates) will be necessary to identify and address any issues promptly and to optimize cleaning cycles.

The selection of membranes will be guided by the results of the ongoing pilot study, in which different membranes are being tested for sulfuric acid separation. Their performance is evaluated based on acid rejection capacity and specifically, the ability to concentrate sulfuric acid (reject) while maintaining very low acidity in the permeate stream at acceptable flow rates. The study also examines the rate at which three different membranes deteriorate under various enrichment stages, i.e., at different sulfuric acid concentrations. Performance monitoring includes tracking the evolution of differential pressure across the membrane, which provides insight into degradation or fouling. Given the nature of this system, additional process safeguards are required, including buffer tanks, standby membrane units, and redundant pumps, to ensure continuous operation and handle variations in flow or equipment maintenance. For further sulfuric acid enrichment, particularly above ~3 wt.% concentration from dilute acid (1-3 wt.%) up to ~96 wt.% is achieved using well-established technologies such as vacuum evaporation.

Optimization of Operating Conditions for the New Process

While pilot tests have demonstrated the feasibility of the hybrid process, optimizing its operating conditions at a commercial scale will be an ongoing challenge. This includes fine-tuning parameters such as:

- Thermal oxidizer temperature and residence time: To ensure complete oxidation of all sulfur species under varying feed conditions.
- Absorption column design and operating parameters: Optimizing packing material, liquid distribution, and L/G ratio for maximum SO₂ absorption efficiency.
- Membrane operating pressure and configuration: To achieve desired concentration factors and permeate quality while minimizing energy consumption and membrane fouling.
- Integration with the Claus unit: Ensuring seamless recycling of concentrated sulfuric acid and managing its impact on the Claus furnace operation.

Advanced process modeling, simulation, and potentially pilot-scale testing will be crucial for optimizing these parameters and developing robust control strategies for commercial implementation. The use of digital twins and AI-driven optimization tools could significantly accelerate this process and enable real-time performance adjustments.

Addressing these technical challenges and operational considerations through careful design, robust material selection, and advanced control strategies is key to ensure the adoption of the novel hybrid water-based absorption and membrane technology for deep desulfurization. The continuous monitoring and optimization throughout the operational life cycle is critical to maintain its advantage against existing and future technologies.

Third-Party Implementation and Operation

The increasing complexity, capital intensity, and specialized expertise required for emission control technologies like the novel hybrid deep desulfurization process often lead industrial operators to consider third-party implementation and operation models. This approach, commonly known as outsourcing or build-own-operate (BOO) / build-own- operate-transfer (BOOT) models, allows companies to focus on their core business while leveraging the specialized capabilities of external providers for non-core but critical functions. For a novel technology such as the hybrid water- based absorption and membrane system, engaging a third party can offer significant strategic and operational advantages.

Business Models for Outsourcing

Several business models can facilitate third-party involvement in the implementation and operation of TGTUs:

- **Build-Own-Operate (BOO):** In a BOO model, a third-party company designs, finances, builds, owns, and operates the TGTU facility. The industrial operator (e.g., refinery or gas plant) typically enters into a longterm service agreement to purchase the treated tail gas or to pay for the sulfur removal service. This model minimizes the capital expenditure for the operator and transfers the operational risks and responsibilities to the third party.
- **Build-Own-Operate-Transfer (BOOT):** Similar to BOO, but with an added transfer clause. After a specified period (e.g., 15-25 years), the ownership of the TGTU facility is transferred to the industrial operator, often at a nominal fee. This model allows the operator to eventually gain full control of the asset while still benefiting from the third party's expertise and reduced upfront investment.
- **Service Contract (Operate & Maintain - O&M):** In this model, the industrial operator owns the TGTU facility, but contracts a third party to operate and maintain it. This is suitable for operators who prefer to retain asset ownership but lack the specialized in-house expertise or resources for optimal operation and maintenance of advanced technologies. The third party is typically paid a service fee based on performance metrics.
- **Engineering, Procurement, and Construction (EPC) with O&M:** A third party is responsible for the entire design, procurement, and construction of the TGTU. Once built, the third party may also provide ongoing operation and maintenance services under a separate contract. This offers a comprehensive solution from project inception to ongoing operation.

Advantages for the Operator

Engaging a third party for the implementation and operation of an advanced TGTU like the hybrid system can provide numerous benefits to the industrial operator:

- **Reduced Capital Expenditure (CAPEX):** For BOO/BOOT models, the third party finances the project, freeing up the operator's capital for core business investments. This is particularly attractive for large-scale projects or companies with limited capital budgets.
- **Access to Specialized Expertise:** Third-party providers often possess deep expertise in designing, building, and operating specific technologies. They have specialized engineers, operators, and

maintenance personnel who are intimately familiar with the nuances of the technology, ensuring optimal performance and troubleshooting capabilities. This is especially valuable for novel or complex processes like the hybrid absorption-membrane system.

- **Risk Transfer:** Operational risks, including technology performance, maintenance, and compliance risks, are largely transferred to the third party. This reduces the operator's exposure to potential technical failures, cost overruns, or regulatory penalties.
- **Focus on Core Business:** By outsourcing non-core activities, the industrial operator can concentrate its resources, management attention, and strategic efforts on its primary production activities, leading to increased efficiency and competitiveness.
- **Guaranteed Performance and Compliance:** Service agreements with third parties often include strict performance guarantees (e.g., sulfur removal efficiency, uptime) and compliance clauses, ensuring that the TGTU consistently meets environmental regulations. Penalties for non-performance incentivize the third party to maintain high standards.
- **Operational Efficiency and Cost Predictability:** Third parties, driven by performance-based contracts, are incentivized to operate the TGTU as efficiently as possible, optimizing energy consumption and chemical usage. This can lead to lower overall operating costs. Furthermore, service fees often provide predictable operational expenses, simplifying budgeting.
- **Faster Project Implementation:** Specialized third parties often have streamlined processes and established supply chains, which can lead to faster project execution and commissioning of the TGTU facility.

Conclusions

The deep desulfurization of Claus unit tail gas is a critical imperative for the oil and gas industry, driven by increasingly stringent environmental regulations and the global commitment to sustainable operations. This paper has introduced a novel hybrid water-based absorption and membrane process, positioning it as a highly promising alternative to conventional amine based Tail Gas Treatment Units (TGTUs).

We have explored the fundamental principles and inherent limitations of the traditional Claus process, highlighting the persistent challenge of residual sulfur compounds in tail gas streams, particularly COS and CS₂. The section on conventional TGTU technologies like reduction/absorption processes underscored the trade-offs between high sulfur recovery efficiency and significant energy consumption, capital costs, and environmental footprint.

The core of this paper detailed the innovative hybrid approach, which combines thermal oxidation of all sulfur species to SO₂, followed by efficient absorption in mildly acidic water. The subsequent concentration of the resulting sulfuric acid using advanced membrane technologies (RO/NF) and its ultimate recycling back to the Claus furnace for decomposition represents a step towards a near-zero waste sulfur recovery process. Pilot tests have successfully validated the feasibility of both the absorption and membrane concentration steps, demonstrating the potential for high efficiency and purity.

The technical, economic, and environmental advantages of this hybrid process are substantial. It eliminates the energy-intensive amine regeneration step, effectively treats problematic COS and CS₂, significantly reduces CO₂ coabsorption, and promotes a closed-loop, zero-waste sulfur recovery system. These benefits translate into lower operating costs, potentially reduced capital expenditure, a significantly smaller carbon footprint, and enhanced compliance with the most stringent emission standards.

While the technology holds promise, successful commercial implementation will require addressing certain technical challenges and operational considerations, such as managing feed gas variability, ensuring long-term membrane durability, and optimizing process control. However, the continuous advancements in materials science, process control, and digital technologies, including AI and digital twins, offer robust solutions to these challenges.

Furthermore, the paper highlighted the growing trend towards third-party implementation and operation models, which can provide industrial operators with access to specialized expertise, reduced capital outlay, and transferred operational risks, thereby facilitating the adoption of such advanced technologies. Looking ahead, the sulfur recovery industry is poised for further transformation through process intensification, carbon capture and utilization, enhanced digitalization, and a stronger embrace of circular economy principles.

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